SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

School of Technology, Design and Computer Application
Bachelor of Technology
Department of Computer Engineering (04)
SUBJECT: Innovation and Design Thinking(2)

Smart Hospital Resource & Appointment Management System

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Abstract

- The project "**Smart Hospital Resource & Appointment Management System**
- " focuses on improving hospital and patient management using online technology.
- It helps patients easily check **bed availability**, **doctor timing**, and **appointment status** from their mobile phones.
- The system gives **real-time updates** to avoid long waiting hours and make hospital operations more efficient.

Introduction

- In today's world, managing healthcare efficiently is very important.
- Many hospitals still use manual systems for patient entry and updates.
- This causes problems like waiting in lines, lack of communication, and delayed updates.
- Our system focuses on **digitalizing healthcare services** to improve patient convenience and hospital management.

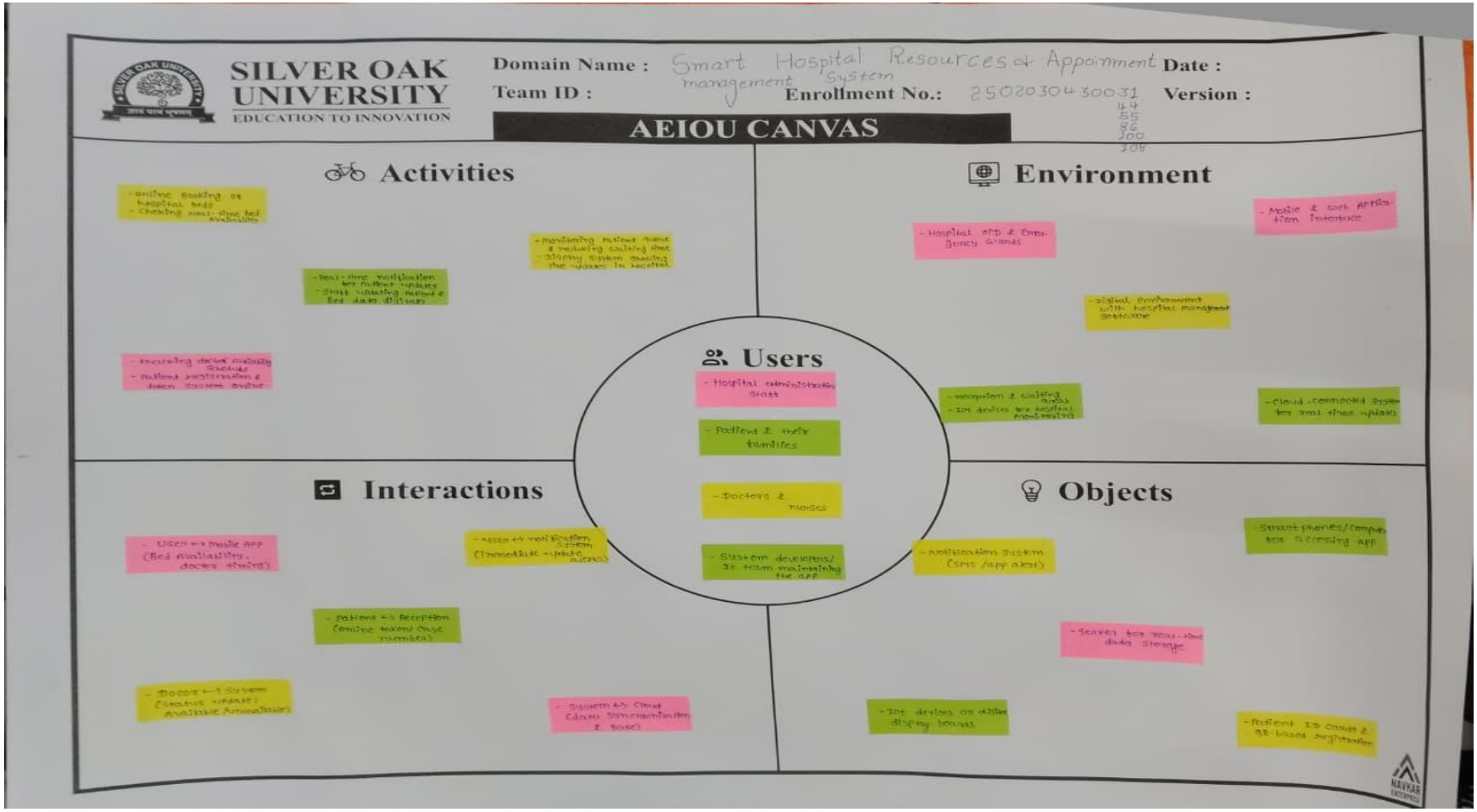
Literature Survey/Research

Sr. No.	Title / Research Work	Technology Used	Key Findings / Outcome
1.	IoT-Based Smart Healthcare System	IoT, Cloud	Real-time health data collection using sensors and remote monitoring of patients.
2.	Online Hospital Management System	Web, Database	Simplifies hospital operations like appointments, billing, and bed management.
3.	*Doctor Appointment & Scheduling Portal*	Android, PHP, MySQL	Helps patients book and track doctor appointments online, reducing waiting time.

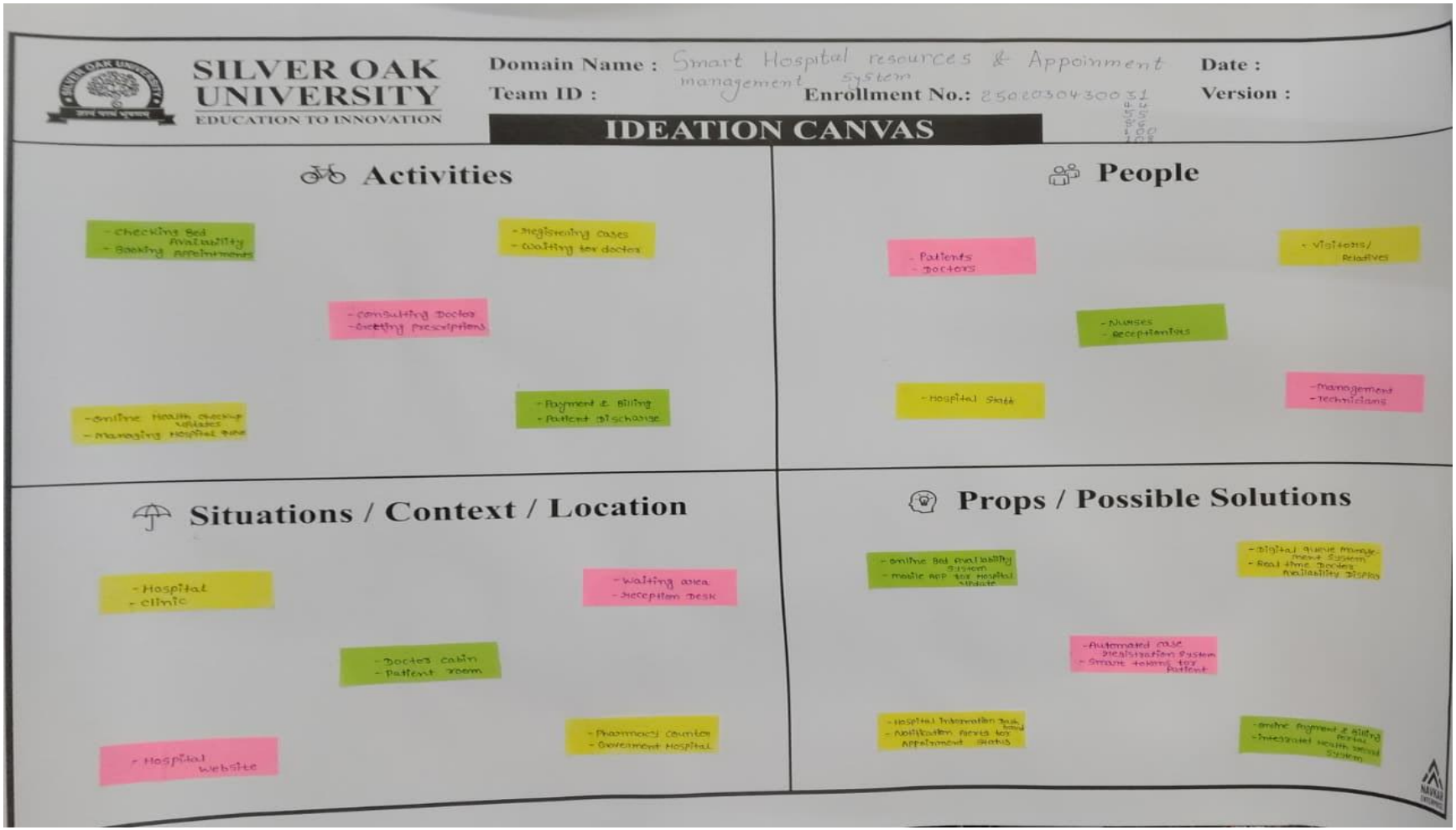
Proposed Work

- Develop an **online system** to show **live bed availability**, **doctor timings**, and **patient queue status**.
- Patients can check updates easily on their **mobile phones**.
- Hospital staff can **update information instantly** through an admin panel.
- Helps reduce waiting time, manage crowds, and improve hospital communication.

AEIOU Canvas



Ideation Canvas



☰ Admin Dashboard

Admin Panel

Home

Add Doctor

Add Staff

Logout

Welcome Admin

Total Doctors: 4

Total Staff: 1

Total Patients: 6

Total Appointments: 5

Total Beds: 117

☰ My Patients

Doctor Panel

Home

My Patients

Logout

Dr. Meena Sharma - Cardiology

Patient	Age	Date	Reason	Status	Prescription	Report	Action
meena	25	2026-04-09	ASS	Pending	Write... Save	View Delete	Done
meena	14	2026-04-09	ASS	Pending	Write... Save	Choose File No file chosen Upload	Done
meena	25	2026-04-09	Ass	Pending	Write... Save	Choose File No file chosen Upload	Done
meena	25	2026-04-09	Ass	Pending	Write... Save	Choose File No file chosen Upload	Done
meena	25	2026-04-09	AS	Pending	Write... Save	Choose File No file chosen Upload	Done

☰ Doctor Schedule

Patient Panel

Home

Book Appointment

My Appointments

Doctor Schedule

Bed Availability

Logout

Doctor Schedule

Doctor	Date	Time
Dr. Raj Patel - Orthopedics	2026-04-07	00:14:00 - 17:00:00
Dr. Priya Nair - Pediatrics	2026-04-07	10:09:00 - 17:25:00
Dr. Meena Sharma - Cardiology	2026-04-25	15:42:00 - 22:41:00

☰ Bed Availability

Patient Panel

Home

Book Appointment

My Appointments

Doctor Schedule

Bed Availability

Logout

Bed Availability

Ward	Total	Occupied	Available	Status
Word no.1	50	18	32	Available
Word no.2	20	8	12	Available
Word no.3	25	13	12	Available
Word no.4	22	16	6	Available

Manage Bed Availability

Staff Panel

Home

Manage Appointments

Manage Beds

Doctor Schedule

Logout

Add Ward / Bed Info

Ward Name

Total Beds

Occupied Beds

Add

All Bed Availability

Ward	Total	Occupied	Available	Status	Action
Word no.1	50	18	32	Available	Discharge | Delete
Word no.2	20	8	12	Available	Discharge | Delete
Word no.3	25	13	12	Available	Discharge | Delete
Word no.4	22	16	6	Available	Discharge | Delete

Doctor Schedule (Date Wise)

Staff Panel

- Home
- Manage Appointments
- Manage Beds
- Doctor Schedule
- Logout

Add Doctor Schedule

Select Doctor | dd-----yyyy | --:-- -- | --:-- -- | Add Schedule

Schedule List

Doctor	Date	Time	Action
Dr. Raj Patel - Orthopedics	2026-04-07	00:14:00 - 17:00:00	Delete
Dr. Priya Nair - Pediatrics	2026-04-07	10:09:00 - 17:25:00	Delete
Dr. Meena Sharma - Cardiology	2026-04-25	15:42:00 - 22:41:00	Delete

☰ Staff Dashboard

Staff Panel

- Home
- Manage Appointments
- Manage Beds
- Doctor Schedule
- Logout

Welcome Staff: staff

Total Beds: 117	Available Beds: 62	Today's Appointments: 0	Total Doctors: 4	Occupancy Rate: 47%

Problem

- **Bed Availability:**
  Patients do not know if beds are available in hospitals until they visit physically.

- **Long Waiting Queues:**
  Patients have to stand outside the hospital for a long time to get their case number or appointment.

- **Doctor Availability:**
  Patients are not informed about which doctors are present or when they will arrive.

- **No Real-time Updates:**
  There is no system to give immediate updates about hospital changes or doctor schedules.

Novelty of Definition

- Provides **real-time hospital data** directly on mobile phones.
- Combines **doctor timing, bed info, and patient queue** in one platform.
- Uses **automatic updates and notifications** for patients and staff.
- Improves both **healthcare access** and **wellness experience** for users.

Solution

- Develop an **online healthcare management system** that shows:
- **Live bed status** on a mobile app or website.
- **Doctor's schedule and availability** in real time.
- **Digital queue system** for patients to take tokens online.
- **Instant notifications** for any update or change in hospital information.
- Helps reduce waiting time and improve hospital transparency.

Work carried out till date

Work Done	Description	Status
Problem Analysis	Studied issues in hospital management	Completed
Design Phase	Planned layout for mobile/web system	Completed
Prototype	Created demo for bed & doctor updates	In Progress
Testing	Collecting sample data for testing	Pending

Future Work & Conclusion

- **Future Work:**
- Add features like online appointment booking and patient report tracking.
- Connect multiple hospitals to one platform for better reach.
- Include AI suggestions for patient wellness and health tips.
- **Conclusion:**
- The system helps hospitals and patients stay connected in real time.
- It reduces crowding, saves time, and ensures better communication.
- A small step toward **smart, digital healthcare and wellness.**

References

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Java: The Complete Reference – Herbert Schildt

Java How to Program – Paul Deitel & Harvey Deitel

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Thank you